

HUFT – DATABASE MANAGEMENT SYSTEM

PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

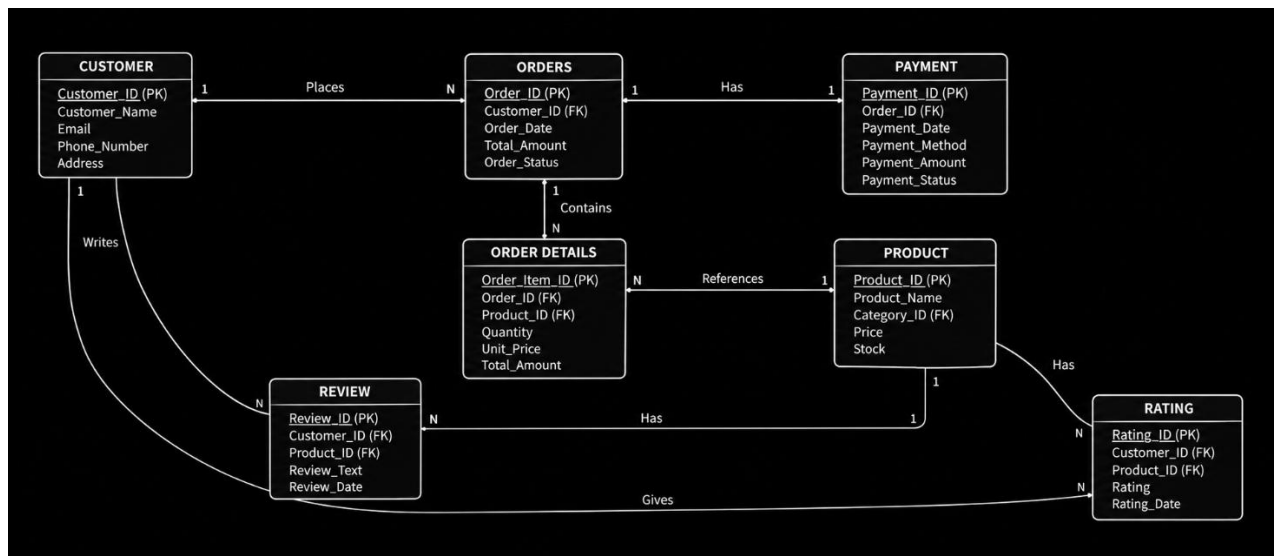
1. REVIEW TABLE

Attribute	Data Type	Description
Review_ID	NUMBER	Unique review ID – Primary Key
Customer_ID	NUMBER	Review submit panna customer – Foreign Key
Product_ID	NUMBER	Product being reviewed
Review_Text	VARCHAR2(200)	Customer's review/comment
Review_Date	DATE	Date of the review

2.RATING TABLE

Column	Data Type	Description
Rating_ID	NUMBER	Unique ID for each rating
Customer_ID	NUMBER	Customer who gives the rating
Product_ID	NUMBER	Product being rated
Rating	NUMBER	Rating given by customer from 1 to 5
Rating_Date	DATE	Date of the rating

3.RELATIONSHIP DIAGRAM



3.TABLE CREATION

3.1 CREATE REVIEW TABLE

```
SQL> CREATE TABLE HuftReview (  
  2     Review_ID NUMBER PRIMARY KEY,  
  3     Customer_ID NUMBER REFERENCES CUST3(CID),  
  4     Product_ID NUMBER REFERENCES HuftProduct(Product_ID),  
  5     Review_Text VARCHAR2(200),  
  6     Review_Date DATE  
  7 );
```

Table created.

```
SQL> DESC HuftReview;
```

Name	Null?	Type
REVIEW_ID	NOT NULL	NUMBER
CUSTOMER_ID		NUMBER
PRODUCT_ID		NUMBER
REVIEW_TEXT		VARCHAR2(200)
REVIEW_DATE		DATE

3.2 CREATE RATING TABLE

```
SQL> CREATE TABLE HuftRating (  
  2     Rating_ID NUMBER PRIMARY KEY,  
  3     Customer_ID NUMBER REFERENCES CUST3(CID),  
  4     Product_ID NUMBER REFERENCES HuftProduct(Product_ID),  
  5     Rating NUMBER(1) CHECK (Rating BETWEEN 1 AND 5),  
  6     Rating_Date DATE  
  7 );
```

Table created.

```
SQL> DESC HuftRating;
```

Name	Null?	Type
RATING_ID	NOT NULL	NUMBER
CUSTOMER_ID		NUMBER
PRODUCT_ID		NUMBER
RATING		NUMBER(1)
RATING_DATE		DATE

4.REVIEW DATA INSERTION

```
SQL> INSERT INTO HuftReview VALUES
  2  (1, 1, 101, 'Excellent dog food', TO_DATE('20-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO HuftReview VALUES
  2  (2, 2, 102, 'Good quality cat food', TO_DATE('21-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO HuftReview VALUES
  2  (3, 3, 103, 'Very good collar', TO_DATE('21-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO HuftReview VALUES
  2  (4, 1, 104, 'Good grooming shampoo', TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO HuftReview VALUES
  2  (5, 2, 106, 'Very useful cat litter', TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.
```

```
SQL> INSERT INTO HuftReview VALUES
  2  (6, 3, 107, 'My dog likes these treats', TO_DATE('23-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

4.1 REVIEW TABLE DISPLAY

```
SQL> SELECT * FROM HuftReview;
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
REVIEW_TEXT
```

```
REVIEW_DA
```

```
1 1 101
Excellent dog food
20-SEP-26
```

```
2 2 102
Good quality cat food
21-SEP-26
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
REVIEW_TEXT
```

```
REVIEW_DA
```

```
3 3 103
Very good collar
21-SEP-26
```

```
4 1 104
Good grooming shampoo
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
REVIEW_TEXT
```

```
REVIEW_DA
```

```
22-SEP-26
5 2 106
Very useful cat litter
22-SEP-26
```

```
6 3 107
```

```
REVIEW_ID CUSTOMER_ID PRODUCT_ID
```

```
REVIEW_TEXT
```

```
REVIEW_DA
```

```
My dog likes these treats
23-SEP-26
```

```
6 rows selected.
```

5.RATING DATA INSERTION

```
SQL> INSERT INTO HuftRating VALUES
  2 (1, 1, 101, 5, TO_DATE('20-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO HuftRating VALUES
  2 (2, 2, 102, 4, TO_DATE('21-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO HuftRating VALUES
  2 (3, 3, 103, 5, TO_DATE('21-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO HuftRating VALUES
  2 (4, 1, 104, 4, TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO HuftRating VALUES
  2 (5, 2, 106, 5, TO_DATE('22-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO HuftRating VALUES
  2 (6, 3, 107, 4, TO_DATE('23-09-2026','DD-MM-YYYY'));

1 row created.
```

5.1 RATING TABLE DISPLAY

```
SQL> SELECT * FROM HuftRating;
```

RATING_ID	CUSTOMER_ID	PRODUCT_ID	RATING	RATING_DA
1	1	101	5	20-SEP-26
2	2	102	4	21-SEP-26
3	3	103	5	21-SEP-26
4	1	104	4	22-SEP-26
5	2	106	5	22-SEP-26
6	3	107	4	23-SEP-26

```
6 rows selected.
```

6 . RETRIEVE PRODUCT REVIEW DETAILS

```
SQL> SELECT
  2     r.Review_ID,
  3     c.CNAME AS Customer_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     r.Review_Text,
  7     r.Review_Date
  8 FROM HuftReview r
  9 JOIN CUST3 c
 10     ON r.Customer_ID = c.CID
 11 JOIN HuftProduct p
 12     ON r.Product_ID = p.Product_ID
 13 ORDER BY r.Review_Date DESC;
```

```
REVIEW_ID CUSTOMER_NAME          PRODUCT_ID
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      6 Ravi                      107
Dental Chew Sticks
My dog likes these treats
23-SEP-26
```

```
REVIEW_ID CUSTOMER_NAME          PRODUCT_ID
-----
PRODUCT_NAME
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
      4 Abi                      104
Pet Grooming Shampoo
Good grooming shampoo
22-SEP-26
```

```
REVIEW_ID CUSTOMER_NAME          PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
REVIEW_TEXT
```

```
-----  
REVIEW_DA
```

```
-----  
          5 Anu                      106  
Cat Litter  
Very useful cat litter  
22-SEP-26
```

```
REVIEW_ID CUSTOMER_NAME          PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
REVIEW_TEXT
```

```
-----  
REVIEW_DA
```

```
-----  
          2 Anu                      102  
Grain Free Cat Food  
Good quality cat food  
21-SEP-26
```

```
REVIEW_ID CUSTOMER_NAME          PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
REVIEW_TEXT
```

```
-----  
REVIEW_DA
```

```
-----  
          3 Ravi                      103  
Leather Dog Collar  
Very good collar  
21-SEP-26
```

```
REVIEW_ID CUSTOMER_NAME          PRODUCT_ID
```

```
-----  
PRODUCT_NAME
```

```
-----  
REVIEW_TEXT
```

```
-----  
REVIEW_DA
```

```
-----  
          1 Abi                      101  
Chicken Dog Food  
Excellent dog food  
20-SEP-26
```

```
6 rows selected.
```

7. CALCULATE AVERAGE PRODUCT RATINGS

```
SQL> SELECT
  2     Product_ID,
  3     AVG(Rating) AS Average_Rating
  4 FROM HuftRating
  5 GROUP BY Product_ID
  6 ORDER BY Product_ID;
```

PRODUCT_ID	AVERAGE_RATING
101	5
102	4
103	5
104	4
106	5
107	4

6 rows selected.

8. FILTER PRODUCTS BY AVERAGE RATING

```
SQL> SELECT
  2     Product_ID,
  3     AVG(Rating) AS Average_Rating
  4 FROM HuftRating
  5 GROUP BY Product_ID
  6 HAVING AVG(Rating) >= 4
  7 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	AVERAGE_RATING
101	5
103	5
106	5
104	4
107	4
102	4

6 rows selected.

9. FILTER PRODUCTS BY AVERAGE RATING WITH PRODUCT NAME

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(r.Rating) AS Average_Rating
  5 FROM HuftRating r
  6 JOIN HuftProduct p
  7     ON r.Product_ID = p.Product_ID
  8 GROUP BY
  9     p.Product_ID,
10     p.Product_Name
11 HAVING AVG(r.Rating) >= 4
12 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
101	Chicken Dog Food	5.00
103	Leather Dog Collar	5.00
106	Cat Litter	5.00
104	Pet Grooming Shampoo	4.00
107	Dental Chew Sticks	4.00
102	Grain Free Cat Food	4.00

6 rows selected.

10. Product Rating Report

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     COUNT(r.Rating_ID) AS Total_Ratings,
  5     ROUND(AVG(r.Rating), 2) AS Average_Rating,
  6     MAX(r.Rating) AS Highest_Rating,
  7     MIN(r.Rating) AS Lowest_Rating
  8 FROM HuftRating r
  9 JOIN HuftProduct p
10     ON r.Product_ID = p.Product_ID
11 GROUP BY p.Product_ID, p.Product_Name
12 ORDER BY Average_Rating DESC;
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS AVERAGE_RATING HIGHEST_RATING LOWEST_RATING
-----
101
Chicken Dog Food
1 5 5 5

103
Leather Dog Collar
1 5 5 5

PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS AVERAGE_RATING HIGHEST_RATING LOWEST_RATING
-----
106
Cat Litter
1 5 5 5

104
Pet Grooming Shampoo

```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS AVERAGE_RATING HIGHEST_RATING LOWEST_RATING
-----
1 4 4 4

107
Dental Chew Sticks
1 4 4 4

102

PRODUCT_ID
-----
PRODUCT_NAME
-----
TOTAL_RATINGS AVERAGE_RATING HIGHEST_RATING LOWEST_RATING
-----
Grain Free Cat Food
1 4 4 4

6 rows selected.

```